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Written Assignment - I

Q.1 Explain the following deep learning terminologies

i) Deep learning

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- Deep learning is transforming the way machines understand, learn and interact with complex data.
- Deep learning mimics neural networks of the human brain, it enables computers to autonomously uncover patterns and make informed decisions from vast amounts of unstructured data.
- Neural network consists of layers of interconnected nodes or neurons, that collaborate to process input data.
- In fully connected deep neural network, data flows through multiple layers, where each neuron performs nonlinear transformation.
- In a deep neural network, the input layer receives data, which passes through hidden layers that transforms the data using non-linear functions.
- The final layer output layer generates the model's prediction.
- Applications :-
  a) Computer Vision
  b) NLP
  c) Speech Recognition
  d) Medical diagnosis.

ii) Deep belief networks

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- Deep Belief Networks (DBNs) are sophisticated artificial neural networks used in the field of DL, a subste of ML.
- They are designed to discover and learn patterns within large sets of data automatically.
- Imagine them as multi-layered networks, where each layer is capable of making sense of the information received from previous one, gradually building up a complex understanding of the overall data.
- DBNs are composed of multiple layers of stochastic, or randomly determined, units.
- These units are known as Restricted Boltzmann Machines (RBMs)
- Each layer in a DBN aims to extract different features from the input data, with lower layers identifying basic patterns and higher layers recognizing more abstract concepts.
- This structure allows DBNs to effectively learn complex representations of data, which makes them particularly useful for tasks like Image and speech recognition, where the input data is high-dimensional and requires a deep level of understanding.

- The architecture of DBN's also makes them good at unsupervised learning, where the goal is to understand and label input data without explicit guidance.
- This characterstic is particularly useful in scenarios where labelled data is scarce or when the goal is to explore the structure of the data without any ~~per~~ preconceived labels.

iii) Boltzmann machines

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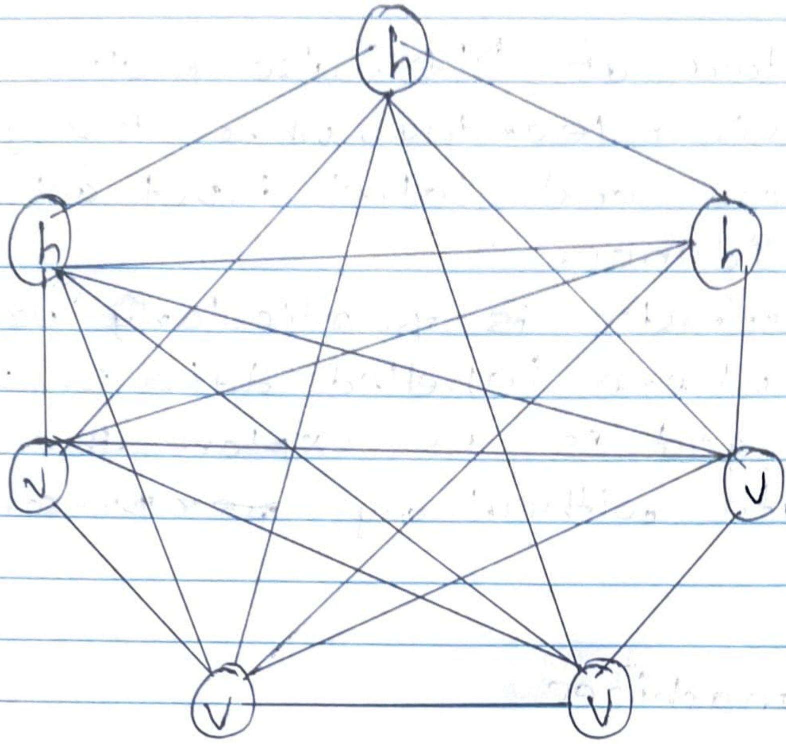
- A ~~deep~~ Boltzmann machine is a type of stochastic neural network that learns patterns from data using energy-based modeling.
- It is primarily used for feature learning, dimensionality reduction and optimization problems
- Characterstics :-

a). Every node in the network is connected to every other node.

b). Neurons update their states probabilistically, rather than using deterministic activation function like ReLU.

c) Uses bidirectional connections for better feature representation.



v - visible nodes
h - hidden nodes

- There are two types of nodes in the Boltzmann machine:
  i) Visible : those nodes which we can and do measure,
  ii) Hidden : those nodes which we cannot or do not measure.
- Although the nodes types are different, the Boltzmann machine considers them as the same and everything works as one single system.
- The training data is fed into the Boltzmann machine and the weights of the system are adjusted accordingly.
- Boltzmann machine helps us understand abnormalities by learning about the working of the system in normal conditions.

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iv) Restricted Boltzmann machine (RBN)
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- The restricted term refers to that we are not allowed to connect the same type layer to each other.
- In other words, the two neurons of the input layer or hidden layer can't connect to each other.
- Although the hidden layer and visible layer can be connected to each other.
- As in this machine, there is no output layer so the question arises how we are going to identify, adjust the weights and how to measure the that our prediction is accurate or not.
- All the questions have one answer, that is Restricted Boltzmann Machine.
- The RBM algorithm was proposed by Geoffrey Hinton, which learns probability distribution over its sample training data inputs.
- It has seen wide applications in different areas of supervised/unsupervised machine learning such as feature learning, dimensionality reduction, classification, collaborative filtering and topic modeling.
- Applications :-

a) Collaborative filtering
b) NLP
c) Anamoly detection

v) Deep neural network (DNN)

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- A deep neural network (DNN) is an artificial neural network with multiple hidden layers between the input and output layers.
- It is a core deep learning model used for complex tasks such as image recognition, NLP and speech recognition.
- A DNN consists of :
a) Input layer - takes raw data
b) Hidden layer - Multiple layers of neurons that extracts patterns from data.
c) Output layer - Produces final predictions.

- Each neuron processes information using the following:

$$y = f(Wx + b)$$

where,
W = weights
x = input
b = bias
f = activation function

- Key Features :-

a) Unlike shallow networks, DNNs learn hierarchical representation of data.

b) Activation functions like ReLU, sigmoid, and tanh allows the model to capture complex patterns.

c) DNNs are trained using backpropagation and optimization techniques like SGD and ADAM.

- Types :-

a) Feedforward (FNN) - Data moves in one direction, from input to output.

b) Convolutional (CNN) - Designed for image processing, using convolutional layers to detect patterns in visual data.

c) Recurrent (RNN) - Used for sequential data like speech and text.

- Applications :-

a) Computer vision

b) NLP

c) Healthcare

d) Speech Recognition.

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F) Deep autoencoder

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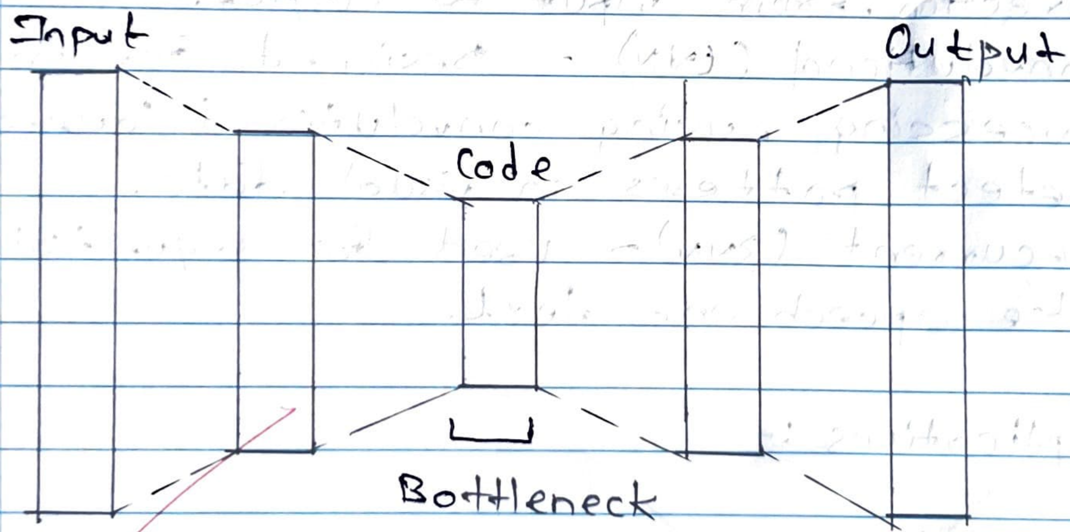
- Autoencoders are simple learning networks that aim to transform inputs into outputs with the minimum possible error.
- They follow encoder-decoder architecture.
- The encoder maps the input to latent space and decoder reconstructs the input.
- The architecture of autoencoder is



- There are three main components.
- Encoder is the first component of network that compresses the input into a latent space representation.
- The encoder encodes the input image as the compressed representation in a reduced dimension.
- Next component is a bottleneck.

- The latent space is the hidden space in which the data lies in the bottleneck.
- The latent space contains a compressed representation of data, sometimes also called as a code.
- The third component is the decoder.
- It decodes the encoded image back to the original dimension by reconstructing the input from the latent space representation.
- It is the only information the decoder is allowed to use to try to reconstruct the input as faithfully as possible.
- TO perform well, the network has to learn to extract the most relevant features in the bottleneck.
- The decoded image is a lossy reconstruction of the original image.

Q.2 Write a short note on 3 classes of deep neural networks.

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i) Feedforward Neural Network (FNN)

=> A FNN is a simplest form of DNN where information moves in one direction - from input to output - without loops or cycles.

- Structure :-

a) Input layer

b) Hidden layer

c) output layer

- Key characterstics :-

a) No memory of past inputs

b) Uses activation function like ReLU, sigmoid

c) Trained using backpropagation and gradient descent.

- Applications :-

a) Image classification

b) stock market prediction

c) Medical diagnosis.

ii) Convolutional Neural Network (CNN)

=> A CNN is designed for processing spatial data, especially images and videos.

- It uses specialized layers to extract features like edges, textures and objects from images.

- Structure :-
  a) Convolutional layers
  b) Pooling layers
  c) Fully connected layers.
- Key characterstics :-
  a) Reduces the number of trained parameters
  b) Detects objects regardless of position in the image.
  c) Early layers detect simple patterns, deeper layers detect complex.
- Applications :-
  a) Image recognition
  b) Object detection
  c) Medical imaging

iii) <u>Recurrent Neural Network (RNN)</u>

→ A RNN is designed for sequential data, where past inputs influences future outputs.
- It has loops that allows information to persist, making it ideal for tasks like speech recognition and language modeling.
- Structure :-
  a) Input layer
  b) Recurrent layer
  c) Output layer
- Key characterstics :-
  a) Maintains context from previous inputs
  b) Adapts to different input sizes.
  c) Gradient issues.

- Applications:-

a) Speech recognition
b) Language translation

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